# GENERAL SCIENCE BOOK

Test Preparation - Notes & MCQs

**PST / JEST - SPSC - Screening Tests**

Matric Level - Sindh Textbook Basics

**Short Notes - Everyday Examples - 199 Practice MCQs with Answer Keys**


## MR ACADEMY

StudySphere

Free study material for the students of Pakistan - learn, practice, prepare, succeed.

# Contents

Chapter 1 - Human Body and Health
Chapter 2 - Plants and Photosynthesis
Chapter 3 - Animals, Habitats and Environment
Chapter 4 - Matter and Materials
Chapter 5 - Force, Motion and Energy
Chapter 6 - Heat and Light
Chapter 7 - Sound and Electricity
Chapter 8 - Earth and Space
Chapter 9 - Water, Air and Pollution
Chapter 10 - Food, Nutrition and Disease Prevention
Mixed Practice Test (All Chapters)
Science Glossary - Alfaaz ke Matlab
Golden Facts Chart
Exam Tips

**How to use this book:** Har chapter mein pehle short notes parhein, phir rozmarra ki misaalein dekhein, aur aakhir mein practice MCQs khud hal karein. Jawab chapter ke aakhir mein diye gaye Answer Key se check karein. Science ratta lagane se nahi, roz ki zindagi se jorne se yaad hoti hai - har baat ko apne aas paas ki cheezon mein dhoondein.

Chapter 1

# Human Body and Health

Insani jism ek hairat angez machine hai - haddiyan, dil, phephre, dimaag sab mil kar kaam karte hain. PST/JEST aur SPSC tests mein jism ke organs, bones aur vitamins par sawal lazmi aate hain. Is chapter mein tamam buniyadi maloomat asaan alfaaz mein di gayi hain.

## Notes - Asaan Alfaaz Mein

- **Haddiyan (Bones):** baligh insaan ke jism mein kul **206 haddiyan** hoti hain (nauzaida bacche mein taqreeban 300). Sab se lambi haddi **femur** hai jo raan (thigh) mein hoti hai, aur sab se chhoti haddi **stapes** hai jo kaan mein hoti hai. Khopri (skull) dimaag ki aur pasliyan (ribs) dil aur phephron ki hifazat karti hain.
- **Pathay (Muscles):** jism mein 600 se zyada pathay hain jo harkat mein madad karte hain. Dil bhi ek khaas patha hai jo kabhi thaka nahi karta - poori zindagi dharkta rehta hai.
- **Dil aur Khoon:** dil khoon ko pure jism mein pump karta hai; dil ke **4 khane (chambers)** hain. Khoon mein teen khaas cells hain - **Red blood cells** oxygen le jate hain, **White blood cells** germs se larte hain, aur **Platelets** khoon ko jamne (clot) mein madad karte hain.
- **Saans ka Nizam:** hawa naak se hokar windpipe (trachea) se phephron (lungs) tak jati hai. Hum oxygen andar lete hain aur carbon dioxide bahar nikalte hain.
- **Hazme ka Nizam:** khana munh se shuru hokar food pipe, phir pait (stomach) aur phir **chhoti aanth (small intestine)** mein jata hai jahan ghiza ke ajza khoon mein jazb hote hain. Bari aanth pani jazb karti hai.
- **Gurde (Kidneys):** khoon ko saaf karke fazla peshab (urine) ki shakal mein bahar nikalte hain. **Jild (skin)** jism ka sab se bara organ hai - pasine ke zariye fazla bahar nikalti hai aur jism ka darja hararat control karti hai.
- **Dimaag aur Ehsaas:** dimaag poore jism ko control karta hai. Paanch senses hain - dekhna (aankh), sunna (kaan), soonghna (naak), chakhna (zuban), chhoona (jild).
- **Jism ki pemaish:** sehatmand jism ka aam darja hararat **37 degree C (98.6 degree F)** hota hai, aur aram ki halat mein nabz (pulse) taqreeban **72 dafa per minute** chalti hai. Baligh insaan ke munh mein kul **32 daant** hote hain.

## Misaalein - Everyday Examples

**Example 1:** Jab hum tez dorte hain to dil ki dharkan aur saans dono tez ho jate hain - kyun?

Dorne se pathon ko zyada energy aur oxygen ki zaroorat parti hai.

Is liye dil zyada khoon pump karta hai aur phephre tez saans lete hain.

Ye jism ka khudkar (automatic) nizam hai - hamein sochna nahi parta.

**Example 2:** Kalai (wrist) par ungli rakh kar nabz ginna

Do ungliyan kalai par halka sa dabayein aur ghari dekh kar ek minute tak dharkanein ginein.

Sehatmand baligh ki nabz aram mein taqreeban 72 per minute hoti hai; warzish ke baad ye barh jati hai.

## Aam Ghaltiyan - Common Mistakes

- Ye samajhna ke dil sirf left side par hota hai - asal mein dil seenay ke darmiyan hota hai, bas us ki nok thori left taraf jhuki hoti hai.

- Ye samajhna ke hum oxygen lete hain aur oxygen hi bahar nikalti hai - bahar nikalne wali hawa mein carbon dioxide zyada hoti hai.
- Ye samajhna ke antibiotic har bimari ki dawa hai - antibiotics sirf bacteria par asar karte hain; zukaam jaise viral marz par nahi.
- Ye samajhna ke haddiyan be-jaan cheez hain - haddiyan zinda tissues hain, in mein khoon ki naliyan hoti hain aur tooti haddi khud jud jati hai.

## Practice MCQs

**1.** How many bones are there in an adult human body?

(a) 206 (b) 306 (c) 106 (d) 226

**2.** The longest bone in the human body is:

(a) Stapes (b) Femur (c) Rib (d) Skull

**3.** The smallest bone in the human body is found in the:

(a) Nose (b) Finger (c) Ear (d) Toe

**4.** The largest organ of the human body is:

(a) Liver (b) Heart (c) Brain (d) Skin

**5.** Which organ pumps blood to all parts of the body?

(a) Kidney (b) Liver (c) Heart (d) Stomach

**6.** Which blood cells carry oxygen?

(a) Red blood cells (b) White blood cells (c) Platelets (d) Nerve cells

**7.** Which blood cells fight against germs?

(a) Red blood cells (b) Platelets (c) Muscle cells (d) White blood cells

**8.** The normal temperature of a healthy human body is about:

(a) 35 degree C (b) 37 degree C (c) 40 degree C (d) 32 degree C

**9.** The normal pulse rate of a healthy adult at rest is about:

(a) 40 beats per minute (b) 72 beats per minute (c) 120 beats per minute (d) 30 beats per minute

**10.** How many teeth does an adult human have?

(a) 28 (b) 30 (c) 36 (d) 32

**11.** The digestion of food begins in the:

(a) Stomach (b) Small intestine (c) Mouth (d) Liver

**12.** Food nutrients are absorbed into the blood in the:

(a) Small intestine (b) Stomach (c) Large intestine (d) Food pipe

**13.** The main work of the kidneys is to:

(a) Pump the blood (b) Digest the food (c) Produce oxygen (d) Filter the blood and remove waste

**14.** Which vitamin is made in our skin with the help of sunlight?

(a) Vitamin A (b) Vitamin D (c) Vitamin C (d) Vitamin B12

**15.** Lack of vitamin C in food causes the disease:

(a) Scurvy (b) Night blindness (c) Rickets (d) Anaemia

**16.** The brain is protected by the:

(a) Ribs (b) Skin (c) Skull (d) Backbone

**17.** How many chambers are there in the human heart?

(a) Four (b) Two (c) Three (d) Five

## Answer Key - Chapter 1

1. (a) 2. (b) 3. (c) 4. (d) 5. (c) 6. (a) 7. (d) 8. (b) 9. (b) 10. (d) 11. (c) 12. (a) 13. (d) 14. (b) 15. (a) 16. (c) 17. (a)

## Try Yourself - Quick Drills

1. Baligh insaan ke jism mein kul kitni haddiyan hoti hain?
2. Jism ka sab se bara organ konsa hai?
3. Sehatmand aadmi ki nabz ek minute mein taqreeban kitni hoti hai?
4. Khoon saaf karne wale organs ke naam likhein.
5. Dhoop se konsa vitamin banta hai?

**Drill Answers:** 1. 206 2. Jild (Skin) 3. 72 dafa 4. Gurde (Kidneys) 5. Vitamin D

Chapter 2

# Plants and Photosynthesis

Pauday zameen par zindagi ki bunyaad hain - ye apna khana khud banate hain aur hamein oxygen dete hain. Photosynthesis, pauday ke hissay aur beej ka ugna (germination) tests ke pakke topics hain.

## Notes - Asaan Alfaaz Mein

- **Photosynthesis:** woh amal jis mein hare pauday sooraj ki roshni, pani aur hawa ki carbon dioxide se apna khana (glucose) banate hain aur **oxygen** bahar chhorte hain. Khana banana din ki roshni mein hota hai.
- **Chlorophyll:** patton mein paya jane wala sabz rang (pigment) jo sooraj ki roshni pakarta hai. Isi liye patte hare hote hain, aur patte ko pauday ki **rasoi (kitchen)** kehte hain.
- **Stomata:** patton mein bareek surakh jin se carbon dioxide andar jati hai, oxygen bahar aati hai aur fazil pani bhi bhaap ban kar nikalta hai.
- **Jadein (Roots):** mitti se pani aur minerals jazb karti hain aur pauday ko mazbooti se jamaye rakhti hain. Gajar ek jarr hi hai jo khana store karti hai.
- **Tana (Stem):** pauday ko sahara deta hai, jaron se pani patton tak aur patton se khana poore pauday tak pohchata hai.
- **Phool aur Beej:** phool pauday ka woh hissa hai jis se beej aur phal bante hain. Shehad ki makkhiyan phoolon ke darmiyan paragan (pollen) le ja kar **pollination** mein madad karti hain.
- **Germination (Beej ka ugna):** beej ko ugne ke liye teen cheezein lazmi hain - **pani, hawa aur munasib garmi**. Pehle chhoti si jarr neeche jati hai, phir ankhua (shoot) upar nikalta hai.
- **Khaas Pauday:** registan ke pauday (cactus) mein kaante asal mein tabdeel shuda patte hain aur mota tana pani store karta hai. **Pitcher plant** kamzor mitti mein rehta hai is liye keeray kha kar apni khorak poori karta hai.

## Misaalein - Everyday Examples

**Example 1:** Do gamle - ek dhoop mein, ek andhere kamre mein

Ek hafta baad dhoop wala gamla hara bhara rahega, andhere wala peela par jayega.

Is se saabit hota hai ke pauday ko khana banane ke liye roshni lazmi hai.

**Example 2:** Bheege hue chane (gram) ki misaal

Khushk chana sakht hota hai; pani mein bhigone se phool jata hai aur 1-2 din mein us mein se safed ankhua nikal aata hai.

Yahi ugne (germination) ki shuruaat hai - pani, hawa aur garmi mili to beej jag utha.

**Example 3:** Khidki ki taraf jhuka hua pauda

Kamre mein rakha gamla hamesha roshni wali taraf jhuk jata hai - pauday roshni ki taraf barhte hain kyunke khana roshni se banta hai.

## Aam Ghaltiyan - Common Mistakes

- Ye samajhna ke pauday sirf din ko saans lete hain - saans (respiration) paudon mein din raat har waqt chalta hai; sirf khana banana (photosynthesis) roshni mein hota hai.
- Ye samajhna ke pauday mitti khate hain - pauday apna khana khud banate hain; mitti se sirf pani aur minerals lete hain.

- Ye samajhna ke har pauday mein phool hote hain - kuch pauday (jaise fern) bina phool ke hote hain aur beej bhi nahi banate.
- Ye samajhna ke cactus ke kaante aise hi hain - asal mein woh tabdeel shuda patte hain jo pani ko bhaap ban kar urne se rokte hain.

## Practice MCQs

**1.** The process by which green plants make their food is called:

(a) Respiration (b) Germination (c) Photosynthesis (d) Pollination

**2.** The green pigment in leaves is called:

(a) Haemoglobin (b) Chlorophyll (c) Cellulose (d) Starch

**3.** Which gas do plants take in for photosynthesis?

(a) Oxygen (b) Nitrogen (c) Hydrogen (d) Carbon dioxide

**4.** Which gas is released by plants during photosynthesis?

(a) Carbon dioxide (b) Nitrogen (c) Smoke (d) Oxygen

**5.** The tiny pores on leaves are called:

(a) Stomata (b) Veins (c) Petals (d) Buds

**6.** Which part of the plant absorbs water and minerals from the soil?

(a) Leaves (b) Flowers (c) Roots (d) Fruits

**7.** Which part carries water from the roots to the leaves?

(a) Flower (b) Stem (c) Seed (d) Fruit

**8.** A seed needs which three things to germinate?

(a) Only water (b) Water, air and suitable warmth (c) Only darkness (d) Only fertilizer

**9.** Bees help plants in:

(a) Germination (b) Respiration (c) Pollination (d) Cutting leaves

**10.** Seeds are formed from the:

(a) Root (b) Leaf (c) Stem (d) Flower

**11.** Which of these is a desert plant?

(a) Cactus (b) Lotus (c) Rose (d) Mango tree

**12.** Which plant eats insects?

(a) Neem tree (b) Sunflower (c) Pitcher plant (d) Grass

**13.** Which plant grows in water?

(a) Cactus (b) Date palm (c) Pine tree (d) Lotus

**14.** The food made by plants is stored in the form of:

(a) Starch (b) Salt (c) Fat only (d) Protein only

**15.** The spines of a cactus are actually its modified:

(a) Flowers (b) Leaves (c) Roots (d) Fruits

**16.** Trees help us mainly by:

(a) Making smoke (b) Drying the rivers (c) Blocking the rain (d) Taking in carbon dioxide and giving out oxygen

**17.** Which part of the plant is called its kitchen?

(a) Root (b) Stem (c) Leaf (d) Flower

## Answer Key - Chapter 2

1. (c) 2. (b) 3. (d) 4. (d) 5. (a) 6. (c) 7. (b) 8. (b) 9. (c) 10. (d) 11. (a) 12. (c) 13. (d) 14. (a) 15. (b) 16. (d) 17. (c)

## Try Yourself - Quick Drills

1. Photosynthesis mein konsi gas bahar nikalti hai?
2. Patton ke hare rang ki wajah konsa pigment hai?
3. Beej ugne ke liye teen lazmi cheezein likhein.
4. Keeray khane wale pauday ka naam likhein.
5. Jarr (root) ke do kaam likhein.

**Drill Answers:** 1. Oxygen 2. Chlorophyll 3. Pani, hawa, munasib garmi 4. Pitcher plant 5. Pani/minerals jazb karna, pauday ko jamaye rakhna

Chapter 3

# Animals, Habitats and Environment

Har janwar apne mahol ke mutabiq bana hai - registan ka oont aur barf ka bhalu is ki behtareen misaalein hain. Janwaron ki qismein, food chain aur mahol ki hifazat tests mein bar bar pooche jate hain.

## Notes - Asaan Alfaaz Mein

- **Habitat:** kisi janwar ya pauday ke rehne ki qudrati jagah ko us ka habitat kehte hain - jaise machhli ka habitat pani, oont ka registan. Janwar apne habitat ke mutabiq dhale hote hain, ise **adaptation** kehte hain.
- **Mammals:** woh janwar jo bacche dete hain, unhein doodh pilate hain aur jism par baal hote hain - gaaye, insaan, billi. **Whael** aur **chamgadar (bat)** bhi mammals hain, chahe ek pani mein rehti hai aur dusra urta hai.
- **Birds:** paron (feathers) wale janwar jo anday dete hain. Magar har parinda ur nahi sakta - **penguin** aur **shutarmurgh (ostrich)** ur nahi sakte. Ostrich sab se bara parinda hai.
- **Fish and others:** machhli pani mein **gills** se saans leti hai aur fins se tairti hai. **Amphibians** (mendak) pani aur zameen dono par rehte hain. **Reptiles** (saanp, chhipkali) rengte hain aur anday dete hain.
- **Insects:** keeray makoron ke **6 pair nahi, 6 taangein** hoti hain - cheunti, titli, makkhi. Makri (spider) ki 8 taangein hoti hain, is liye woh insect nahi hai.
- **Khane ki Aadat:** sirf pauday khane wale **herbivores** (gaaye, hiran), sirf gosht khane wale **carnivores** (sher, cheetah), aur dono khane wale **omnivores** (insaan, kauwa) kehlate hain.
- **Food Chain:** qudrat mein khane ka silsila - ghaas ko hiran khata hai, hiran ko sher. Ghaas banane wala (producer) hai, khane wale consumers hain, aur murda cheezon ko mitti mein milane wale bacteria aur fungi **decomposers** hain.
- **Khaas Baatein:** girgit (chameleon) dushman se bachne ke liye rang badalta hai (camouflage). Kuch janwar sardiyon mein lambi neend sote hain (hibernation). Jo janwar khatam ho jayein unhein extinct kehte hain - jaise dodo aur dinosaur.

## Misaalein - Everyday Examples

**Example 1:** Registan mein oont (camel) ki banawat

Us ke kohaan (hump) mein charbi jama hoti hai jo bhooke waqt energy deti hai.

Woh kam pani pi kar kayi din guzar sakta hai aur lambay pair garam ret se jism ko door rakhte hain.

**Example 2:** Ek sada food chain

Dana (grain) ko chooha khata hai, choohe ko saanp, saanp ko uqaab (eagle).

Agar saanp khatam ho jayein to choohe barh kar fasal tabah kar denge - yahi food chain ka balance hai.

## Aam Ghaltiyan - Common Mistakes

- Ye samajhna ke chamgadar (bat) parinda hai - woh mammal hai kyunke bacche deta hai aur doodh pilata hai, anday nahi deta.

- Ye samajhna ke whale machhli hai - whale mammal hai, woh lungs se saans leti hai aur hawa ke liye pani ki satah par aati hai.
- Ye samajhna ke makri (spider) insect hai - us ki 8 taangein hoti hain, insect ki hamesha 6 hoti hain.
- Ye samajhna ke oont ke kohaan mein pani bhara hota hai - asal mein us mein charbi (fat) hoti hai jo energy deti hai.

## Practice MCQs

**1.** The natural home of an animal is called its:

(a) Adaptation (b) Habitat (c) Camouflage (d) Migration

**2.** Which of the following is a mammal?

(a) Bat (b) Owl (c) Eagle (d) Crow

**3.** The largest animal in the world is:

(a) Elephant (b) Blue whale (c) Giraffe (d) Horse

**4.** Which bird cannot fly?

(a) Penguin (b) Sparrow (c) Pigeon (d) Parrot

**5.** The largest bird in the world is:

(a) Eagle (b) Peacock (c) Crane (d) Ostrich

**6.** A fish breathes through its:

(a) Lungs (b) Skin (c) Gills (d) Nose

**7.** A frog is an example of:

(a) Amphibian (b) Reptile (c) Mammal (d) Insect

**8.** How many legs does an insect have?

(a) Eight (b) Four (c) Ten (d) Six

**9.** Which of these has eight legs and is NOT an insect?

(a) Ant (b) Butterfly (c) Spider (d) Housefly

**10.** The cow is a:

(a) Carnivore (b) Herbivore (c) Omnivore (d) Scavenger

**11.** The lion is a:

(a) Herbivore (b) Omnivore (c) Carnivore (d) Decomposer

**12.** Human beings are:

(a) Herbivores (b) Omnivores (c) Carnivores (d) Producers

**13.** The long winter sleep of some animals is called:

(a) Hibernation (b) Migration (c) Camouflage (d) Adaptation

**14.** Changing colour to hide from enemies is called:

(a) Hibernation (b) Migration (c) Pollination (d) Camouflage

**15.** Which of these animals is now extinct?

(a) Dodo (b) Tiger (c) Deer (d) Panda

**16.** In a food chain, grass is called a:

(a) Consumer (b) Producer (c) Decomposer (d) Predator

**17.** The hump of a camel stores:

(a) Water (b) Milk (c) Fat (d) Sand

## Answer Key - Chapter 3

1. (b) 2. (a) 3. (b) 4. (a) 5. (d) 6. (c) 7. (a) 8. (d) 9. (c) 10. (b) 11. (c) 12. (b) 13. (a) 14. (d) 15. (a) 16. (b) 17. (c)

## Try Yourself - Quick Drills

1. Chamgadar (bat) kis group ka janwar hai?
2. Mendak (frog) kis group mein aata hai?
3. Machhli saans kis cheez se leti hai?
4. Sardiyon ki lambi neend ko kya kehte hain?
5. Ek food chain ki misaal likhein.

**Drill Answers:** 1. Mammal 2. Amphibian 3. Gills se 4. Hibernation 5. Ghaas, hiran, sher

Chapter 4

# Matter and Materials

Hamare aas paas har cheez madde (matter) se bani hai. Madde ki teen halatein, elements aur mixtures, aur tezab (acid) aur base ki pehchan - ye science ka sab se buniyadi chapter hai.

## Notes - Asaan Alfaaz Mein

- **Madda (Matter):** har woh cheez jis ka wazan (mass) ho aur jagah gherti ho, madda kehlata hai. Madde ki teen halatein hain - **thos (solid)**: shakl aur hajm pakka (pathar, lakri); **mae (liquid)**: hajm pakka magar shakl bartan ki (pani, doodh); **gas**: na shakl pakki na hajm (hawa, bhaap).
- **Halat ki Tabdeeli:** barf garmi se pighalti hai (melting), pani ubal kar bhaap banta hai (evaporation), bhaap thandi satah par pani ban jati hai (condensation). Barf jamne aur pani ke jamne ka darja **0 degree C** hai, aur pani **100 degree C** par ubalta hai.
- **Elements:** sab se sada madday jo aur sada nahi ho sakte - jaise oxygen, hydrogen, loha (iron), sona (gold), taanba (copper), carbon. Duniya mein 118 ke qareeb elements maloom hain.
- **Compounds:** jab do ya zyada elements chemical tareeqe se mil jayein - pani (H2O) hydrogen aur oxygen ka compound hai, namak aur carbon dioxide bhi compounds hain.
- **Mixtures:** jab cheezein sirf mil jayein, chemical tor par na juren - hawa, samandar ka pani, sherbet. Mixtures ko asaan tareeqon se alag kiya ja sakta hai.
- **Alag Karne ke Tareeqe:** chhanni se (sieving), filter se (filtration - chai ki channi isi ki misaal hai), aur evaporation se - samandar ke pani ko kyaaron mein bhar kar dhoop se namak hasil kiya jata hai.
- **Acids and Bases:** acids ka zaiqa khatta hota hai - lemon, sirka (vinegar), dahi. Bases karwe aur sabun jaisi chiknahat wale hote hain - sabun, toothpaste. **Litmus paper** se pehchan hoti hai: acid mein neela litmus **surkh** ho jata hai, base mein surkh litmus neela ho jata hai.
- **Dhaten (Metals):** chamakti hain aur garmi aur bijli ko achi tarah guzarti hain (taanbe ke taar isi liye bante hain). Ghair-dhaten (non-metals) jaise gandhak (sulphur) aur koyla aisa nahi karte. Maqnaatees (magnet) sirf lohe, nickel aur cobalt ko khenchta hai.

## Misaalein - Everyday Examples

**Example 1:** Band dhakkan wale bartan mein ubalta pani

Dhakkan uthayein to us ke neeche pani ki boondein hongi.

Bartan se uthi bhaap thande dhakkan se takra kar dobara pani ban gayi - yahi condensation hai.

**Example 2:** Samandar ke pani se namak

Khaaron (salt pans) mein samandar ka pani bhar kar dhoop mein chor dete hain.

Pani evaporation se ur jata hai aur peeche safed namak reh jata hai.

**Example 3:** Geelay kapre dhoop mein sookhna

Kapron ka pani bhaap ban kar hawa mein ur jata hai - ye bhi evaporation hai.

Taiz dhoop aur hawa mein evaporation aur tez ho jata hai, is liye kapre jaldi sookhte hain.

## Aam Ghaltiyan - Common Mistakes

- Ye samajhna ke jo safed cheez bartan se nikalti hai wohi asal bhaap hai - asal steam (gas) nazar nahi aati; dikhne wali safed cheez thandi hawa se bani pani ki chhoti boondein hain.

- Ye samajhna ke har chamakti cheez metal hoti hai - sheesha aur heera (diamond) bhi chamakte hain magar metal nahi hain.
- Ye samajhna ke namak pani mein ghul kar khatam ho jata hai - woh ghula hota hai; pani evaporation se ura dein to namak wapas mil jata hai.
- Ye samajhna ke har acid khatarnak hota hai - khane peene wale acids (lemon, dahi) mehfooz hain; khatarnak sirf tez (strong) acids hote hain jo lab mein ehtiyat se istemal hote hain.

## Practice MCQs

**1.** Matter is anything that has mass and occupies:

(a) Colour (b) Smell (c) Light (d) Space

**2.** Which state of matter has a fixed shape and fixed volume?

(a) Liquid (b) Gas (c) Solid (d) Vapour

**3.** Which state of matter takes the shape of its container but has a fixed volume?

(a) Liquid (b) Gas (c) Stone (d) Wood

**4.** At what temperature does ice melt and water freeze?

(a) 100 degree C (b) 50 degree C (c) 37 degree C (d) 0 degree C

**5.** At what temperature does water boil?

(a) 0 degree C (b) 100 degree C (c) 60 degree C (d) 90 degree C

**6.** Steam changing back into water is called:

(a) Evaporation (b) Condensation (c) Melting (d) Freezing

**7.** Water is a compound of:

(a) Carbon and oxygen (b) Oxygen and nitrogen (c) Hydrogen and carbon (d) Hydrogen and oxygen

**8.** Which of the following is an element?

(a) Water (b) Salt (c) Iron (d) Air

**9.** Air is an example of a:

(a) Mixture (b) Element (c) Compound (d) Metal

**10.** Lemon juice is sour because it contains an:

(a) Base (b) Sugar (c) Oil (d) Acid

**11.** In an acid, blue litmus paper turns:

(a) Green (b) Red (c) Yellow (d) Black

**12.** Common salt is obtained from sea water by:

(a) Evaporation (b) Filtration (c) Using a magnet (d) Freezing

**13.** Which metal is mostly used for making electric wires?

(a) Sulphur (b) Coal (c) Copper (d) Wood

**14.** A magnet attracts:

(a) Iron (b) Wood (c) Rubber (d) Plastic

**15.** Which of these dissolves in water?

(a) Sand (b) Stones (c) Sugar (d) Oil

**16.** Sand can be separated from water by:

(a) A magnet (b) Filtration (c) Freezing it (d) Melting it

**17.** Burning of wood is a:

(a) Physical change (b) Melting (c) Freezing (d) Chemical change

## Answer Key - Chapter 4

1. (d) 2. (c) 3. (a) 4. (d) 5. (b) 6. (b) 7. (d) 8. (c) 9. (a) 10. (d) 11. (b) 12. (a) 13. (c) 14. (a) 15. (c) 16. (b) 17. (d)

## Try Yourself - Quick Drills

1. Pani ki teen halaton ke naam likhein.
2. H2O kis cheez ka formula hai?
3. Khatti cheezon mein kya paya jata hai?
4. Chai chhanne ke amal ko kya kehte hain?
5. Taanba (copper) bijli ke taaron mein kyun istemal hota hai?

**Drill Answers:** 1. Barf (solid), pani (liquid), bhaap (gas) 2. Pani (water) 3. Acid (tezab) 4. Filtration 5. Woh bijli ka acha conductor hai

Chapter 5

# Force, Motion and Energy

Dhakka dena, khenchna, gaari ki raftaar aur bijli ki energy - ye sab physics ki bunyaad hai. Force, friction, gravity, speed aur simple machines par tests mein paake sawal aate hain.

## Notes - Asaan Alfaaz Mein

- **Force:** dhakka (push) ya khenchna (pull) force kehlata hai. Force se cheez chalti hai, rukti hai, tez ya sust hoti hai, simt (direction) badalti hai aur naram cheez ki shakl bhi badal sakti hai (mitti dabana).
- **Friction:** do satahon ke darmiyan ragad ki woh taqat jo harkat ki mukhalifat karti hai. Hamwaar (smooth) satah par friction kam aur khurduri (rough) satah par zyada hota hai. Friction ke fayde bhi hain - is ke baghair hum chal nahi sakte aur gaarion ke brake nahi lag sakte.
- **Gravity:** zameen har cheez ko apni taraf khenchti hai, is liye cheezein neeche girti hain. Newton ke bare mein mashhoor hai ke seb girte dekh kar unhon ne gravity ka khayal socha.
- **Speed (Raftar):** speed = faasla (distance) taqseem waqt (time). Misal: gaari 60 km/h se chale to matlab hai ek ghante mein 60 kilometre tay karegi.
- **Simple Machines:** kaam asaan karne wale sada aalaat - **lever** (qainchi, sarriya, see-saw), **pulley** (kunwein ki ghirni jis se balti asani se upar aati hai), **inclined plane** (dhalwan ya ramp jis par bojh charhana asaan hota hai) aur **wheel** (pahiya).
- **Energy:** kaam karne ki salahiyat ko energy kehte hain. Zameen ke liye energy ka sab se bara zariya **sooraj** hai.
- **Energy ki Qismein:** chalti hui cheez mein **kinetic energy** hoti hai; bandh ke ruke hue pani aur khinchi hui rabri (rubber band) mein stored yani **potential energy** hoti hai. Is ke ilawa heat, light, sound aur electrical energy bhi hain.
- **Energy ki Tabdeeli:** energy khatam nahi hoti, sirf roop badalti hai - bulb mein bijli roshni aur garmi ban jati hai, cell (battery) mein chemical energy bijli ban jati hai. Sooraj, hawa aur pani **renewable** zariya hain; koyla aur petrol khatam hone wale (non-renewable) hain.

## Misaalein - Everyday Examples

**Example 1:** Speed ka hisaab - ek sada sawal

Ek bus 120 kilometre ka faasla 2 ghante mein tay karti hai.

Speed = 120 taqseem 2 = 60 kilometre per hour.

Matlab: har ghante mein woh 60 kilometre ja rahi hai.

**Example 2:** Kunwein se balti nikalna

Ghirni (pulley) ki madad se rassi neeche khenchne par balti upar aati hai.

Pulley force ki simt badal deti hai aur kaam asaan ho jata hai.

**Example 3:** Cricket ball ko bat se marna

Bat ka lagna ball par force hai - ball ki raftaar aur simt dono badal jati hain.

Yahi force ki pehchan hai: harkat mein tabdeeli.

## Aam Ghaltiyan - Common Mistakes

- Ye samajhna ke ruki hui cheez par koi force nahi lagta - gravity har waqt har cheez par lagti rehti hai, khari gaari par bhi.

- Ye samajhna ke friction hamesha bura hai - friction na ho to pair phislenge, gaari ruk nahi payegi aur hum qalam bhi nahi pakar sakenge.
- Ye samajhna ke energy khatam ho jati hai - energy kabhi khatam nahi hoti, sirf ek roop se doosre roop mein badal jati hai.
- Ye samajhna ke bhari cheez hamesha halki cheez se pehle girti hai - girne ki raftaar wazan par nahi hoti; farq aksar hawa ki rukawat se parta hai.

## Practice MCQs

**1.** A push or a pull acting on a body is called:

(a) Energy (b) Speed (c) Friction (d) Force

**2.** Force applied on soft clay can change its:

(a) Shape (b) Colour (c) Smell (d) Taste

**3.** The force that opposes motion between two surfaces is:

(a) Gravity (b) Magnetism (c) Friction (d) Electricity

**4.** The pulling force of the Earth is called:

(a) Friction (b) Gravity (c) Pressure (d) Sound

**5.** Speed is calculated by:

(a) Time divided by distance (b) Distance divided by time (c) Distance multiplied by time (d) Distance plus time

**6.** A car travels 100 kilometres in 2 hours. Its speed is:

(a) 200 km/h (b) 25 km/h (c) 50 km/h (d) 100 km/h

**7.** Which of the following is a simple machine?

(a) Television (b) Computer (c) Radio (d) Pulley

**8.** A sloping ramp is an example of:

(a) Inclined plane (b) Wheel (c) Pulley (d) Scissors

**9.** A pair of scissors works as a:

(a) Pulley (b) Wedge (c) Lever (d) Screw

**10.** A moving object has which kind of energy?

(a) Potential energy (b) Chemical energy (c) No energy (d) Kinetic energy

**11.** A stretched rubber band has:

(a) Potential energy (b) Kinetic energy (c) Sound energy (d) Light energy

**12.** An electric bulb changes electrical energy into:

(a) Sound only (b) Light and heat (c) Wind (d) Chemical energy

**13.** The biggest source of energy for the Earth is the:

(a) Moon (b) Coal (c) Petrol (d) Sun

**14.** Which of these is a renewable source of energy?

(a) Coal (b) Petrol (c) Solar energy (d) Diesel

**15.** The to-and-fro motion of a swing (jhoola) is called:

(a) Circular motion (b) Oscillatory motion (c) Straight line motion (d) Random motion

**16.** Friction helps us in:

(a) Walking without slipping (b) Seeing things (c) Hearing sounds (d) Breathing

**17.** Which surface has the least friction?

(a) Sandpaper (b) Ice (c) Road (d) Grass field

## Answer Key - Chapter 5

1. (d) 2. (a) 3. (c) 4. (b) 5. (b) 6. (c) 7. (d) 8. (a) 9. (c) 10. (d) 11. (a) 12. (b) 13. (d) 14. (c) 15. (b) 16. (a) 17. (b)

## Try Yourself - Quick Drills

1. Force ki saada definition likhein.
2. Speed ka formula likhein.
3. Kunwein ki ghirni kis machine ki misaal hai?
4. Chalti gaari mein konsi energy hoti hai?
5. Zameen ki kheenchne wali taqat ko kya kehte hain?

**Drill Answers:** 1. Dhakka (push) ya khenchna (pull) 2. Speed = Distance / Time 3. Pulley 4. Kinetic energy 5. Gravity

Chapter 6

# Heat and Light

Garmi ka behna aur roshni ka safar - dono hamari roz ki zindagi ka hissa hain. Thermometer, conduction, aaina, saaya aur rainbow tests ke mashhoor topics hain.

## Notes - Asaan Alfaaz Mein

- **Heat and Temperature:** heat ek energy hai jo garam cheez se thandi cheez ki taraf behti hai. Temperature garmi ki pemaish hai jo **thermometer** se napi jati hai; is ki akai degree Celsius hai. Sehatmand jism ka darja 37 degree C hota hai.
- **Heat ke Behne ke teen Tareeqe: Conduction** - thos cheezon mein (garam chai mein rakha chamcha garam ho jana). **Convection** - mae aur gaso mein; garam hissa halka hokar upar uthta hai (pani garam karne par). **Radiation** - bina kisi cheez ke zariye; sooraj ki garmi khala se guzar kar ham tak isi se pohanchti hai.
- **Conductors and Insulators:** loha aur taanba garmi ke ache conductor hain; lakri, plastic aur oon (wool) insulator hain. Sardiyon mein ooni kapre garmi nahi dete - woh jism ki garmi ko bahar jane se rokte hain.
- **Phailna (Expansion):** dhaten garmi pa kar phailti hain. Railway ki patriyon ke darmiyan is liye halka sa faasla rakha jata hai, aur thermometer ka para ya rangdar pani bhi garam hokar phailta hai to upar charhta hai.
- **Roshni ka Safar:** roshni hamesha **seedhi lakeer** mein safar karti hai aur is ki raftaar sab se zyada hai - taqreeban **300,000 kilometre per second**.
- **Reflection (Roshni ka Palatna):** aaine (mirror) se roshni palat kar wapas aati hai, is liye hamein apna chehra dikhta hai. Hamwaar satah se barabar reflection hota hai, khurduri satah se bikhra hua.
- **Refraction (Roshni ka Jhukna):** jab roshni ek cheez se doosri mein jati hai (hawa se pani mein) to jhuk jati hai. Isi liye pani ke glass mein rakhi chamach turi hui ya tedi lagti hai.
- **Saaya aur Rang:** jo cheez roshni ko ruk de (opaque, jaise lakri) us ka saaya banta hai; sheesha transparent hai - roshni guzar jati hai; oiled paper translucent hai - kuch roshni guzarti hai. Safed roshni saat (7) rangon se mil kar banti hai - prism aur barish ke baad ke **rainbow** mein ye saat rang alag nazar aate hain.

## Misaalein - Everyday Examples

**Example 1:** Garam chai ka chamcha garam ho jana

Dhat ke chamche ka woh hissa jo haath mein hai, thodi der mein garam mehsus hota hai.

Garmi chai se chamche ke zariye haath tak ayi - ye conduction hai.

**Example 2:** Thali mein rakhi cheez aur dhoop wali gaari

Dhoop mein khari kaali gaari safed gaari se zyada garam ho jati hai.

Kaala rang roshni aur garmi zyada jazb (absorb) karta hai - is liye garmiyon mein halke rang ke kapre pehne jate hain.

**Example 3:** Sooraj ki garmi ham tak kaise pohanchti hai

Zameen aur sooraj ke darmiyan khala (khaali jagah) hai - wahan na conduction ho sakta hai na convection.

Garmi radiation ke zariye aati hai, jo bina kisi madde ke safar karti hai.

## Aam Ghaltiyan - Common Mistakes

- Ye samajhna ke ooni kapra jism ko garmi deta hai - woh asal mein jism ki apni garmi ko bahar jane se rokta hai (insulator hai).
- Ye samajhna ke saaya hamesha hota hi rehta hai - saaya sirf tab banta hai jab koi opaque cheez roshni ke raste mein aaye.
- Ye samajhna ke metal ka bartan thanda hota hai is liye us ka temperature kam hota hai - woh haath ki garmi tezi se khenchta hai (acha conductor hai), is liye thanda lagta hai.
- Ye samajhna ke rainbow mein teen rang hote hain - safed roshni ke saat rang hote hain jo rainbow mein dikhte hain.

## Practice MCQs

**1.** Temperature is measured with a:

(a) Thermometer (b) Barometer (c) Speedometer (d) Ammeter

**2.** Heat always flows from:

(a) A cold body to a hot body (b) Nowhere (c) Only upward (d) A hot body to a cold body

**3.** Transfer of heat in solids mainly takes place by:

(a) Convection (b) Radiation (c) Conduction (d) Evaporation

**4.** Heat from the Sun reaches the Earth by:

(a) Radiation (b) Conduction (c) Convection (d) Friction

**5.** Which of the following is a good conductor of heat?

(a) Wood (b) Plastic (c) Rubber (d) Iron

**6.** Woollen clothes keep us warm because wool is:

(a) A good conductor of heat (b) A source of heat (c) A bad conductor of heat (d) Heavier than cotton

**7.** Light always travels in a:

(a) Curved line (b) Straight line (c) Zigzag line (d) Circle

**8.** The speed of light is about:

(a) 300 km per second (b) 3,000 km per second (c) 300,000 km per second (d) 30 km per second

**9.** The bending of light when it passes from air into water is called:

(a) Reflection (b) Refraction (c) Radiation (d) Conduction

**10.** The bouncing back of light from a mirror is called:

(a) Reflection (b) Refraction (c) Dispersion (d) Convection

**11.** Glass is an example of a _____ material.

(a) Opaque (b) Dark (c) Magnetic (d) Transparent

**12.** A shadow is formed when light is blocked by:

(a) An opaque object (b) Clean water (c) Thin air (d) A mirror

**13.** White light is made of how many colours?

(a) Three (b) Seven (c) Five (d) Ten

**14.** A rainbow is formed when sunlight passes through:

(a) Dust only (b) Smoke (c) Raindrops (d) Clouds only

**15.** In a thermometer, the liquid rises when heated because it:

(a) Contracts (b) Freezes (c) Turns into gas (d) Expands

**16.** Dark-coloured clothes feel hotter in the sun because they _____ heat more.

(a) Reflect (b) Produce (c) Absorb (d) Ignore

**17.** Small gaps are left between railway tracks because metals _____ when heated.

(a) Expand (b) Contract (c) Melt (d) Vanish

## Answer Key - Chapter 6

1. (a) 2. (d) 3. (c) 4. (a) 5. (d) 6. (c) 7. (b) 8. (c) 9. (b) 10. (a) 11. (d) 12. (a) 13. (b) 14. (c) 15. (d) 16. (c) 17. (a)

## Try Yourself - Quick Drills

1. Temperature napne wale aale ka naam likhein.
2. Roshni ki raftaar kitni hai?
3. Aaine se roshni ke palatne ko kya kehte hain?
4. Saaya kab banta hai?
5. Garmiyon mein halke rang ke kapre kyun pehne jate hain?

**Drill Answers:** 1. Thermometer 2. Taqreeban 300,000 km per second 3. Reflection 4. Jab opaque cheez roshni ko rok de 5. Woh garmi kam jazb karte hain

Chapter 7

# Sound and Electricity

Awaaz tharthrahat (vibration) se banti hai aur bijli band circuit mein behti hai. Echo, lightning, fuse aur conductors - ye do chhote topics tests mein aksar number dilate hain.

## Notes - Asaan Alfaaz Mein

- **Sound kya hai:** har awaaz kisi cheez ke **tharhrane (vibration)** se paida hoti hai - bolte waqt hamara gala tharhrata hai, guitar ke taar tharhrate hain, table par bajta tabla tharhrata hai.
- **Medium ki Zaroorat:** awaaz ko safar ke liye koi madda chahiye - hawa, pani ya thos. **Khala (vacuum) mein awaaz safar nahi kar sakti**, is liye chaand par khamoshi hai.
- **Awaaz ki Raftaar:** hawa mein taqreeban **340 metre per second**. Roshni is se kahin zyada tez hai - is liye bijli ki chamak pehle dikhti hai aur badal ki garaj (thunder) baad mein sunai deti hai.
- **Pitch and Loudness:** awaaz ka ooncha ya neecha hona pitch kehlata hai (bareek awaaz - high pitch, bhari awaaz - low pitch). Awaaz ka tez ya aahista hona loudness hai.
- **Echo (Goonj):** jab awaaz kisi deewar ya pahar se takra kar wapas aaye to goonj sunai deti hai. Ye awaaz ka reflection hai.
- **Electric Circuit:** bijli sirf band (mukammal) raste mein behti hai. Ek circuit mein cell ya battery, taarein, bulb aur switch hote hain. Switch band ho to current behta hai, khula ho to ruk jata hai.
- **Conductors and Insulators:** taanba (copper) bijli ko guzarta hai - conductor hai. Rabar (rubber) aur plastic rookte hain - insulator hain; is liye bijli ke taaron par plastic charha hota hai taake jhatka na lage.
- **Cell, Battery and Fuse:** cell chemical energy ko bijli mein badalta hai; do ya zyada cells mil kar battery kehlate hain. **Fuse** ek hifazati taar hai - current zyada ho jaye to woh pighal kar circuit tor deta hai aur aag lagne se bachata hai. Asmani bijli (lightning) bhi bijli hi ki ek bari shakal hai.

## Misaalein - Everyday Examples

**Example 1:** Bijli chamak aur garaj mein waqfa

Door asman mein pehle chamak dikhti hai, kuch second baad garaj sunai deti hai.

Dono ek saath banti hain, magar roshni awaaz se boht tez hai.

Andaaza: har 3 second ke waqfe ka matlab hai bijli taqreeban 1 kilometre door giri.

**Example 2:** Train ki patri (rail) par kaan lagana

Door se aati train ki awaaz patri ke lohe se safar karke hawa se pehle sunai de sakti hai.

Awaaz thos cheezon mein gaso ke muqable zyada tezi se safar karti hai.

**Example 3:** Torch ka circuit

Torch mein cell, taar, bulb aur button (switch) hai.

Button dabane se circuit mukammal hota hai aur bulb jal uthta hai; button chhorne se circuit toot jata hai.

## Aam Ghaltiyan - Common Mistakes

- Ye samajhna ke awaaz khala (space) mein bhi sunai de sakti hai - awaaz ko medium lazmi chahiye, khala mein koi awaaz nahi ho sakti.
- Ye samajhna ke bijli ki chamak aur garaj alag alag waqt par banti hain - dono ek hi lamhe banti hain; farq sirf raftaar ka hai.

- Ye samajhna ke plastic ke taar se bijli guzar sakti hai - plastic insulator hai, is liye taaron ki hifazati teh bhi isi ki banti hai.
- Ye samajhna ke switch khula ho tab bhi bulb jal sakta hai - circuit adhoora ho to current bilkul nahi behta.

## Practice MCQs

**1.** Sound is produced by:

(a) Light rays (b) Heat waves (c) Shadows (d) Vibrations

**2.** Sound cannot travel through:

(a) Air (b) Vacuum (c) Water (d) Wood

**3.** The speed of sound in air is about:

(a) 340 km/s (b) 340 m/s (c) 34 m/s (d) 3,000 m/s

**4.** We see lightning before we hear thunder because:

(a) Sound travels faster than light (b) Thunder is produced later (c) Clouds stop the sound (d) Light travels faster than sound

**5.** The reflection of sound is called:

(a) Pitch (b) Noise (c) Echo (d) Vibration

**6.** The highness or lowness of a sound is called its:

(a) Pitch (b) Loudness (c) Echo (d) Speed

**7.** The sound of a guitar is produced by its vibrating:

(a) Wooden box only (b) Keys (c) Drum skin (d) Strings

**8.** The complete path through which electric current flows is called a:

(a) Switch (b) Circuit (c) Fuse (d) Cell

**9.** Which material is a good conductor of electricity?

(a) Copper (b) Rubber (c) Plastic (d) Wood

**10.** Electric wires are covered with plastic because plastic is an:

(a) Conductor (b) Magnet (c) Insulator (d) Fuse

**11.** The safety device that melts and breaks the circuit when current is too high is a:

(a) Fuse (b) Bulb (c) Cell (d) Switch

**12.** A battery is made of:

(a) One wire (b) Many bulbs (c) Two or more cells (d) Many switches

**13.** A cell changes chemical energy into:

(a) Sound energy (b) Electrical energy (c) Wind energy (d) Light only

**14.** When a switch is opened, it:

(a) Completes the circuit (b) Charges the cell (c) Stores the current (d) Breaks the circuit

**15.** Sound travels fastest through:

(a) Gases (b) Liquids only (c) Vacuum (d) Solids

**16.** Lightning in the sky is a form of:

(a) Electricity (b) Rain water (c) Heat (d) Wind

**17.** In a torch, the bulb finally gives out:

(a) Sound energy (b) Chemical energy (c) Light energy (d) Wind energy

## Answer Key - Chapter 7

1. (d) 2. (b) 3. (b) 4. (d) 5. (c) 6. (a) 7. (d) 8. (b) 9. (a) 10. (c) 11. (a) 12. (c) 13. (b) 14. (d) 15. (d) 16. (a) 17. (c)

## Try Yourself - Quick Drills

1. Awaaz kis cheez se paida hoti hai?
2. Khala mein awaaz kyun nahi ho sakti?
3. Goonj (echo) kya hoti hai?
4. Bijli ke taaron par plastic kyun charha hota hai?
5. Circuit ko on/off karne wala purza konsa hai?

**Drill Answers:** 1. Vibration (tharathrahat) se 2. Safar ke liye medium nahi hota 3. Awaaz ka takra kar wapas ana 4. Plastic insulator hai, jhatka nahi lagta 5. Switch

Chapter 8

# Earth and Space

Sooraj, aath sayyare, chaand aur zameen ki gardish - ye woh topics hain jin se har screening test mein sawal aata hai. Din raat aur mausamon ki asal wajah samajhna sab se ahem hai.

## Notes - Asaan Alfaaz Mein

- **Solar System:** sooraj darmiyan mein hai aur us ke gird **8 sayyare** ghumte hain - Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune. Mercury sooraj ke sab se qareeb aur Neptune sab se door hai.
- **Mashhoor Sayyare: Jupiter** sab se bara sayyara hai. **Venus** sab se garam aur sab se chamakdar hai - use subah ya shaam ka tara kehte hain, magar asal mein woh sayyara hai. **Mars** ko laal sayyara (red planet) kehte hain aur **Saturn** ke gird khoobsurat halqe (rings) hain.
- **Hamari Zameen:** ye wahid maloom sayyara hai jahan zindagi hai, kyunke yahan pani, hawa aur munasib temperature teeno mojood hain.
- **Rotation (Gardish):** zameen apne mehwar (axis) ke gird 24 ghante mein ek chakkar poora karti hai - isi se **din aur raat** bante hain. Jo hissa sooraj ke samne ho wahan din, doosri taraf raat.
- **Revolution aur Mausam:** zameen sooraj ke gird ek chakkar **365¼ din** mein poora karti hai - ye hamara ek saal hai. Mausam zameen ke mehwar ke **23.5 degree jhukao (tilt)** ki wajah se badalte hain: jis hisse ki taraf jhukao ho wahan dhoop seedhi parti hai aur garmi hoti hai.
- **Chaand (Moon):** zameen ka wahid qudrati satellite hai jo taqreeban 384,000 kilometre door hai. Chaand ki **apni roshni nahi** - woh sooraj ki roshni reflect karta hai. Sirf roshan hissa nazar aane ki wajah se us ki shaklein badalti dikhti hain (phases); poora cycle taqreeban 29.5 din ka hota hai.
- **Eclipses (Grahan):** jab chaand sooraj aur zameen ke darmiyan aa jaye to **solar eclipse (suraj grahan)** hota hai. Jab zameen darmiyan aa jaye aur us ka saaya chaand par pare to **lunar eclipse (chaand grahan)** hota hai.
- **Sitare aur Sayyare:** sitaron ki apni roshni hoti hai - sooraj bhi ek tara (star) hai, zameen ke sab se qareeb wala. Sayyaron ki apni roshni nahi hoti, woh sooraj ki roshni reflect karke chamakte hain. Door ki cheezein dekhne ke liye **telescope** istemal hoti hai.

## Misaalein - Everyday Examples

**Example 1:** Torch aur globe se din raat

Andhere kamre mein globe ke ek taraf torch jalayein - roshan hisse par din, doosri taraf raat hogi.

Globe ko aahista ghumayein to roshan hissa badalta rahega - bilkul asli zameen ki tarah.

**Example 2:** Mausamon ki misaal - June aur December

June mein Pakistan mein sakht garmi hoti hai kyunke hamara hissa sooraj ki taraf jhuka hota hai aur dhoop seedhi parti hai.

December mein wohi hissa door jhuk jata hai, dhoop terhi parti hai aur sardi ho jati hai.

Farq sooraj ke faasle ka nahi - mehwar ke jhukao ka hai.

## Aam Ghaltiyan - Common Mistakes

- Ye samajhna ke mausam is liye badalte hain ke zameen kabhi sooraj ke qareeb aur kabhi door hoti hai - asal wajah mehwar ka 23.5 degree jhukao hai.
- Ye samajhna ke chaand ki apni roshni hoti hai - chaand sirf sooraj ki roshni reflect karta hai.
- Ye samajhna ke Venus tara (star) hai kyunke use subah ka tara kehte hain - woh ek sayyara hai.
- Ye samajhna ke sooraj mashriq se maghrib tak safar karta hai - asal mein zameen ghoomti hai, is liye sooraj hamein chalta nazar aata hai.

## Practice MCQs

**1.** How many planets are there in our solar system?

(a) 7 (b) 8 (c) 9 (d) 10

**2.** The planet nearest to the Sun is:

(a) Venus (b) Mercury (c) Earth (d) Mars

**3.** The largest planet of the solar system is:

(a) Saturn (b) Neptune (c) Jupiter (d) Earth

**4.** Which planet is called the Red Planet?

(a) Venus (b) Jupiter (c) Mercury (d) Mars

**5.** The planet famous for its beautiful rings is:

(a) Saturn (b) Mars (c) Venus (d) Earth

**6.** The brightest planet, also called the morning star, is:

(a) Mars (b) Jupiter (c) Venus (d) Mercury

**7.** Life is known to exist only on:

(a) Mars (b) Venus (c) Jupiter (d) Earth

**8.** The natural satellite of the Earth is:

(a) Moon (b) Sun (c) Mars (d) Venus

**9.** The Moon shines because it _____ sunlight.

(a) Produces (b) Reflects (c) Absorbs fully (d) Creates

**10.** Day and night on Earth are caused by its:

(a) Revolution around the Sun (b) Movement of the Moon (c) Clouds in the sky (d) Rotation on its axis

**11.** The Earth completes one rotation in:

(a) 12 hours (b) 365 days (c) 24 hours (d) 30 days

**12.** Seasons on Earth are caused mainly by:

(a) Changes in distance only (b) The tilt of the axis of the Earth (c) The Moon (d) Strong winds

**13.** The Earth completes one revolution around the Sun in:

(a) 365 and a quarter days (b) 30 days (c) 24 hours (d) 12 hours

**14.** In a solar eclipse, which body comes between the Sun and the Earth?

(a) Mars (b) The Moon (c) Venus (d) A comet

**15.** In a lunar eclipse, the shadow of the Earth falls on the:

(a) Moon (b) Sun (c) Mars (d) Venus

**16.** The Sun is a:

(a) Planet (b) Moon (c) Comet (d) Star

**17.** The phases of the Moon complete their cycle in about:

(a) 7 days (b) 15 days (c) 29.5 days (d) 90 days

## Answer Key - Chapter 8

1. (b) 2. (b) 3. (c) 4. (d) 5. (a) 6. (c) 7. (d) 8. (a) 9. (b) 10. (d) 11. (c) 12. (b) 13. (a) 14. (b) 15. (a) 16. (d) 17. (c)

## Try Yourself - Quick Drills

1. Sab se bara sayyara konsa hai?
2. Laal sayyara kis ko kehte hain?
3. Din raat kis wajah se hote hain?
4. Suraj grahan mein sooraj aur zameen ke darmiyan kya aata hai?
5. Sooraj ek kya hai - sayyara ya tara?

**Drill Answers:** 1. Jupiter 2. Mars ko 3. Zameen ki apne mehwar par gardish (rotation) se 4. Chaand 5. Tara (star)

Chapter 9

# Water, Air and Pollution

Pani aur hawa zindagi ke liye lazmi hain - aur insaan ki laparwahi inhein ganda kar rahi hai. Water cycle, hawa ki tarkeeb aur pollution ki rokaam tests ke ahem topics hain.

## Notes - Asaan Alfaaz Mein

- **Pani:** zameen ki satah ka taqreeban **71 feesad** hissa pani se dhaka hai. Pani ke zariya hain - barish, darya, aur zameen ke andar ka pani (kunein aur tube well).
- **Water Cycle:** sooraj ki garmi se samandar aur daryaon ka pani bhaap banta hai (evaporation), upar ja kar thanda hokar badal banta hai (condensation), phir barish ki shakal mein wapas zameen par aata hai - ye silsila chalta rehta hai.
- **Saaf Pani:** gande pani mein germs hote hain jin se haiiza (cholera) aur typhoid jaisi bimariyan phelti hain. Pani ko **ubalne** se germs mar jate hain aur filter karne se kachra alag ho jata hai.
- **Hawa ki Tarkeeb:** hawa gases ka mixture hai - **Nitrogen taqreeban 78 feesad**, **Oxygen taqreeban 21 feesad**, aur baqi taqreeban 1 feesad mein carbon dioxide aur doosri gases hain.
- **Oxygen ka Kirdar:** oxygen saans lene aur aag jalne ke liye lazmi hai. Hum saans mein oxygen lete hain aur carbon dioxide bahar nikalte hain; pauday carbon dioxide le kar oxygen dete hain - yun qudrat ka balance qaim rehta hai.
- **Air Pollution:** gaarion aur factorion ka dhuwan, kachra jalana aur dhool hawa ko ganda karte hain. Is se khansi, damma (asthma) aur ankhon ki jalan jaisi takleefein hoti hain. Sardiyon mein dhuwan aur dhund mil kar **smog** ban jati hai.
- **Water Pollution:** factorion ka ganda pani aur sheher ka sewerage daryaon mein milne se pani zehreela ho jata hai - machliyan aur doosre jandaar marte hain aur woh pani peene ke qabil nahi rehta. Plastic ki theliyaan mitti mein saalon tak nahi galtin (rot nahi hotin).
- **Hifazat ke Tareeqe:** zyada darakht lagana, kachra muqarrar jagah par phainkna, daryaon mein ganda pani na dalna, plastic ka istemal kam karna aur ghar mein pani zaya na karna. Ozone ki teh zameen ke gird hamein sooraj ki nuqsandeh kirnon se bachati hai.

## Misaalein - Everyday Examples

**Example 1:** Barish ke baad khadda (puddle) gayab ho jana

Subah ka bhara khadda shaam tak sookh jata hai - pani dhoop ki garmi se evaporation se ur gaya.

Wohi pani kabhi badal ban kar kahin aur barish banega - yahi water cycle hai.

**Example 2:** Sheher aur gaon ki hawa ka farq

Barre shehron mein gaarion ke dhuein ki wajah se hawa bhari aur gandi hoti hai; gaon mein darakhton ki wajah se saaf.

Isi liye shehron mein darakht lagana sehat ke liye zaroori hai.

## Aam Ghaltiyan - Common Mistakes

- Ye samajhna ke hawa mein sab se zyada gas oxygen hai - asal mein sab se zyada nitrogen hai (taqreeban 78 feesad), oxygen sirf 21 feesad hai.
- Ye samajhna ke darya mein ganda pani dalne se darya khud saaf kar deta hai - pollution barhta hi hai, jandaar marte hain aur pani peene ke qabil nahi rehta.

- Ye samajhna ke plastic mitti mein jaldi gal jata hai - plastic saalon tak nahi galta, is liye woh mahol ke liye sab se nuqsandeh kachra hai.
- Ye samajhna ke barish ka har pani bina safai ke peene ke liye saaf hota hai - chhat aur hawa ki dhool se pehli barish ka pani ganda hota hai; saaf bartan mein jama karke ubalna behtar hai.

## Practice MCQs

**1.** About what percentage of the surface of the Earth is covered with water?

(a) About 71 percent (b) About 50 percent (c) About 29 percent (d) About 90 percent

**2.** In the water cycle, water vapour changing into clouds is called:

(a) Evaporation (b) Freezing (c) Melting (d) Condensation

**3.** Water rises from seas and rivers as vapour by:

(a) Condensation (b) Pollution (c) Evaporation (d) Filtration

**4.** Boiling drinking water helps to:

(a) Add minerals to it (b) Kill the germs in it (c) Make it dirty (d) Change its colour

**5.** Which disease spreads through dirty water?

(a) Malaria (b) Diabetes (c) Cholera (d) Tooth decay

**6.** The most abundant gas in the air is:

(a) Oxygen (b) Nitrogen (c) Carbon dioxide (d) Hydrogen

**7.** The percentage of oxygen in the air is about:

(a) 21 percent (b) 78 percent (c) 50 percent (d) 1 percent

**8.** Which gas is necessary for burning?

(a) Nitrogen (b) Carbon dioxide (c) Hydrogen (d) Oxygen

**9.** In breathing, human beings give out more of which gas?

(a) Carbon dioxide (b) Oxygen (c) Nitrogen (d) Helium

**10.** The main cause of air pollution in big cities is:

(a) Rainfall (b) Smoke from vehicles and factories (c) Green trees (d) River water

**11.** Smog is a harmful mixture of:

(a) Rain and dust (b) Snow and fog (c) Smoke and fog (d) Steam and air

**12.** Throwing factory waste into rivers mainly causes:

(a) Cleaner air (b) More fish (c) Fertile soil (d) Water pollution

**13.** Plastic bags are harmful for the environment because they:

(a) Rot within a day (b) Feed the animals (c) Do not rot in the soil for years (d) Clean the water

**14.** Which action helps to reduce air pollution?

(a) Planting more trees (b) Burning garbage (c) Cutting forests (d) Making more smoke

**15.** Which layer in the sky protects us from the harmful rays of the Sun?

(a) Cloud layer (b) Dust layer (c) Rain layer (d) Ozone layer

**16.** Which gas do green plants absorb from the air?

(a) Nitrogen (b) Carbon dioxide (c) Oxygen (d) Smoke

**17.** Saving clean water is important because clean water is:

(a) Unlimited (b) Limited (c) Made by taps daily (d) Found equally everywhere

## Answer Key - Chapter 9

1. (a) 2. (d) 3. (c) 4. (b) 5. (c) 6. (b) 7. (a) 8. (d) 9. (a) 10. (b) 11. (c) 12. (d) 13. (c) 14. (a) 15. (d) 16. (b) 17. (b)

## Try Yourself - Quick Drills

1. Hawa mein sab se zyada konsi gas hai?
2. Pani ubalne se kya faida hota hai?
3. Water cycle ke teen marhalay likhein.
4. Pollution kam karne ka ek asaan tareeqa likhein.
5. Darya mein factory ka ganda pani dalna konsa pollution hai?

**Drill Answers:** 1. Nitrogen (taqreeban 78 feesad) 2. Germs mar jate hain 3. Evaporation, condensation, barish (rain) 4. Darakht lagana ya kachra na jalana 5. Water pollution

Chapter 10

# Food, Nutrition and Disease Prevention

Khaana sirf pait bharne ke liye nahi - ye energy, nashonuma aur bimariyon se hifazat teeno deta hai. Food ke ajza, vitamins ki kami ki bimariyan aur bachao ke tareeqe tests mein pakke aate hain.

## Notes - Asaan Alfaaz Mein

- **Khane ke Ajza:** hamare khane mein paanch khaas cheezein honi chahiyein - carbohydrates, proteins, fats, vitamins aur minerals. Sath mein pani aur roughage (resha, sabzi aur phalon mein) bhi lazmi hai.
- **Carbohydrates:** jism ko fauri energy dete hain - roti, chawal, aaloo, cheeni, makai. Khelne aur kaam karne ki taqat inhi se milti hai.
- **Proteins:** jism ki nashonuma (growth) aur toot phoot ki marammat karte hain - gosht, anda, doodh, machhli aur daalein protein ke ache zariya hain. Bacchon ke liye protein khaas zaroori hai.
- **Fats:** tail, ghee aur makkhan energy store karte hain, magar in ki zyadti motapa aur dil ki bimariyon ka sabab banti hai - is liye miqdar munasib ho.
- **Vitamins aur Minerals: Vitamin A** ankhon ke liye (gajar, aam) - kami se raat ko dikhai kam ho jata hai (night blindness). **Vitamin C** (santra, lemon, amla) zakham bharne mein madad karta hai - kami se scurvy hota hai. **Vitamin D aur Calcium** (doodh, dhoop) haddiyon ko mazboot karte hain - kami se bacchon ki haddiyan naram par jati hain (rickets). **Iron** (palak, gosht) khoon banata hai - kami se anaemia (khoon ki kami) hota hai. **Iodine** wala namak goitre (galay ki gilti) se bachata hai.
- **Balanced Diet:** woh khana jis mein tamam ajza munasib miqdar mein hon - roti, daal ya gosht, sabzi, doodh ya dahi aur phal. Din mein 6 se 8 glass saaf pani bhi peena chahiye.
- **Bimariyan Kaise Phelti Hain:** machhar ke kaatne se malaria aur dengue, gande pani aur khane se cholera, typhoid aur hepatitis, hawa ke zariye zukaam aur TB (tuberculosis), aur pagal kutte ke kaatne se rabies.
- **Bachao ke Tareeqe:** waqt par vaccination (teekay), khana khane se pehle aur toilet ke baad sabun se haath dhona, khana dhak kar rakhna, machhar daani ka istemal, aas paas safai, roz daant saaf karna, aur puri neend aur warzish.

## Misaalein - Everyday Examples

**Example 1:** Ek balanced thaali ki misaal

Roti (carbohydrate) + daal (protein) + sabzi (vitamins aur roughage) + dahi (calcium) + ek phal (vitamin C).

Is ek thaali mein tamam ajza munasib miqdar mein mojood hain - yahi balanced diet hai.

**Example 2:** School mein sust bacha - anaemia ki misaal

Ek bacha har waqt thaka rehta hai, rang peela hai - doctor khoon test karke batata hai ke iron ki kami hai (anaemia).

Palak, gosht aur khajoor jaisi iron wali ghiza aur doctor ki dawa se woh theek ho jata hai.

## Aam Ghaltiyan - Common Mistakes

- Ye samajhna ke sirf gosht khana sehatmand hai - jism ko her juz ki zaroorat hai; balanced diet ke baghair koi ek cheez kaafi nahi.
- Ye samajhna ke vitamin ki goliyan khane ki jagah le leti hain - goliyan madadgar ho sakti hain, magar qudrati khana hamesha behtareen hai.
- Ye samajhna ke har bukhaar malaria hota hai - bukhaar boht si wajah se hota hai; khoon ke test se hi asal wajah maloom hoti hai.
- Ye samajhna ke aam zukaam (common cold) antibiotic se theek hota hai - zukaam virus se hota hai, antibiotic us par asar nahi karti; aaram aur pani behtareen tareeqa hai.

## Practice MCQs

**1.** The main energy-giving foods are:

(a) Proteins only (b) Vitamins only (c) Minerals only (d) Carbohydrates

**2.** Foods that build the body and repair it are rich in:

(a) Carbohydrates (b) Fats only (c) Proteins (d) Water

**3.** Which food is a rich source of protein?

(a) Eggs (b) Rice (c) Potato (d) Sugar

**4.** Vitamin A is necessary for healthy:

(a) Bones only (b) Teeth only (c) Hair (d) Eyes

**5.** Lack of vitamin A in diet causes:

(a) Scurvy (b) Night blindness (c) Rickets (d) Goitre

**6.** Deficiency of vitamin C causes the disease:

(a) Scurvy (b) Night blindness (c) Anaemia (d) Rickets

**7.** Which fruit is rich in vitamin C?

(a) Banana only (b) Potato (c) Orange (d) Onion

**8.** Vitamin D and calcium are needed for strong:

(a) Bones (b) Eyes (c) Skin (d) Blood

**9.** Goitre disease is prevented by using salt that contains:

(a) Iron (b) Calcium (c) Iodine (d) Sugar

**10.** Lack of iron in the body causes:

(a) Goitre (b) Anaemia (c) Scurvy (d) Night blindness

**11.** Malaria spreads by the bite of a:

(a) Housefly (b) Rat (c) Dog (d) Mosquito

**12.** Rabies spreads by the bite of an infected:

(a) Mosquito (b) Housefly (c) Cat only (d) Dog

**13.** The best habit to stop germs before eating is to:

(a) Wash hands with soap (b) Rinse hands with dust (c) Wipe hands on a dirty cloth (d) Skip washing

**14.** Vaccination protects our body from:

(a) Hunger (b) Cold weather (c) Diseases (d) Accidents

**15.** Eating too much sugar mainly damages the:

(a) Eyes (b) Teeth (c) Ears (d) Bones

**16.** A balanced diet means eating:

(a) Only meat (b) All food groups in right amounts (c) Only fruits (d) As much as possible

**17.** Roughage (fibre) in food helps in:

(a) Seeing better (b) Hearing better (c) Digestion (d) Growing hair

## Answer Key - Chapter 10

1. (d) 2. (c) 3. (a) 4. (d) 5. (b) 6. (a) 7. (c) 8. (a) 9. (c) 10. (b) 11. (d) 12. (d) 13. (a) 14. (c) 15. (b) 16. (b) 17. (c)

## Try Yourself - Quick Drills

1. Protein ke do ache zariya likhein.
2. Ankhon ke liye konsa vitamin zaroori hai?
3. Haath dhona kab kab lazmi hai?
4. Malaria kis ke kaatne se phelta hai?
5. Balanced diet se kya murad hai?

**Drill Answers:** 1. Anda aur daalein (ya doodh, gosht) 2. Vitamin A 3. Khana khane se pehle aur toilet ke baad 4. Machhar ke kaatne se 5. Tamam food ajza munasib miqdar mein

Final Test

# Mixed Practice Test

Ye test tamam chapters se mila jula hai - bilkul asal test ki tarah. Waqt note karein aur khud ko check karein.

**1.** Which gas is released by green plants during photosynthesis?

(a) Carbon dioxide (b) Nitrogen (c) Smoke (d) Oxygen

**2.** The longest bone in the human body is:

(a) Femur (b) Stapes (c) Rib (d) Skull

**3.** Water boils at a temperature of:

(a) 0 degree C (b) 50 degree C (c) 100 degree C (d) 37 degree C

**4.** The largest planet in the solar system is:

(a) Saturn (b) Mars (c) Earth (d) Jupiter

**5.** Which gas makes up about 78 percent of the air?

(a) Nitrogen (b) Oxygen (c) Carbon dioxide (d) Hydrogen

**6.** Sound cannot travel through:

(a) Air (b) Vacuum (c) Water (d) Iron

**7.** Scurvy is caused by the lack of:

(a) Vitamin A (b) Vitamin D (c) Iron (d) Vitamin C

**8.** A fish breathes with the help of its:

(a) Lungs (b) Skin (c) Gills (d) Fins

**9.** The force that opposes motion is:

(a) Gravity (b) Friction (c) Magnetism (d) Electricity

**10.** The speed of light is about:

(a) 300,000 km per second (b) 340 m/s (c) 300 km per second (d) 1,000 km per second

**11.** The Moon is a satellite of the:

(a) Sun (b) Earth (c) Mars (d) Venus

**12.** Which of these foods is rich in protein?

(a) Lentils (daal) (b) Rice (c) Sugar (d) Oil

**13.** Blue litmus paper turns _____ in an acid.

(a) Green (b) Yellow (c) Black (d) Red

**14.** The hump of a camel stores:

(a) Water (b) Milk (c) Fat (d) Sand

**15.** Dew drops on grass on a winter morning are formed by:

(a) Condensation (b) Evaporation (c) Melting (d) Freezing

**16.** How many chambers are there in the human heart?

(a) Two (b) Three (c) Five (d) Four

**17.** Seasons change on Earth mainly because of:

(a) The distance of the Moon (b) Strong winds (c) The tilt of the axis of the Earth (d) Ocean waves

**18.** Chlorophyll is present in the:

(a) Roots (b) Leaves (c) Flowers (d) Seeds

**19.** Cholera spreads mainly through:

(a) Mosquito bite (b) Dog bite (c) Dirty water (d) Cold air

**20.** Which of these is an electrical insulator?

(a) Copper (b) Rubber (c) Iron (d) Aluminium

**21.** Which planet is nearest to the Sun?

(a) Mercury (b) Venus (c) Earth (d) Mars

**22.** An insect has how many legs?

(a) Eight (b) Four (c) Ten (d) Six

**23.** Salt can be separated from sea water by:

(a) Evaporation (b) A magnet (c) Freezing (d) Filtration only

**24.** Energy stored in a stretched rubber band is:

(a) Kinetic energy (b) Potential energy (c) Sound energy (d) Light energy

**25.** Rabies is spread by the bite of an infected:

(a) Mosquito (b) Housefly (c) Dog (d) Rat

**26.** A seed needs which of these to germinate?

(a) Only water (b) Only soil (c) Only sunlight (d) Water, air and suitable warmth

**27.** The ozone layer protects us from:

(a) Heavy rain (b) Cold winds (c) Harmful rays of the Sun (d) Earthquakes

**28.** The normal resting pulse rate of an adult is about:

(a) 72 beats per minute (b) 40 beats per minute (c) 100 beats per minute (d) 120 beats per minute

**29.** The pitch of a sound tells us whether the sound is:

(a) Loud or soft (b) Near or far (c) Fast or slow (d) High or low

## Answer Key - Mixed Practice Test

1. (d) 2. (a) 3. (c) 4. (d) 5. (a) 6. (b) 7. (d) 8. (c) 9. (b) 10. (a) 11. (b) 12. (a) 13. (d) 14. (c) 15. (a) 16. (d) 17. (c) 18. (b) 19. (c) 20. (b) 21. (a) 22. (d) 23. (a) 24. (b) 25. (c) 26. (d) 27. (c) 28. (a) 29. (d)

Word List

# Science Glossary - Alfaaz ke Matlab

Is list mein is book ke tamam aham science alfaaz ke asaan matlab diye gaye hain. Jo lafz samajh na aaye, yahan se dekh lein.

- **Photosynthesis:** woh amal jismein hare pauday roshni se apna khana banate hain.
- **Chlorophyll:** patton ka sabz rang jo roshni pakarta hai.
- **Germination:** beej ka ugna - pani, hawa aur garmi se shuru hota hai.
- **Pollination:** phoolon mein pollen ka muntaqil hona jis se beej banta hai.
- **Habitat:** kisi jandaar ke rehne ki qudrati jagah.
- **Adaptation:** janwar ya pauday ka apne mahol ke mutabiq dhalna.
- **Food Chain:** khane ka qudrati silsila - ghaas, hiran, sher.
- **Mammal:** bacche dene aur doodh pilane wala janwar - gaaye, whale, chamgadar.
- **Amphibian:** pani aur zameen dono par rehne wala janwar - mendak.
- **Matter:** har cheez jis ka wazan ho aur jagah gherti ho.
- **Evaporation:** mae (liquid) ka bhaap (gas) mein badalna.
- **Condensation:** bhaap ka thanda hokar dobara pani ban jana.
- **Element:** sab se sada madda - oxygen, loha, sona.
- **Compound:** do ya zyada elements ke milne se bani cheez - pani (H2O).
- **Force:** dhakka (push) ya khenchna (pull).
- **Friction:** do satahon ki ragad jo harkat ki mukhalifat karti hai.
- **Gravity:** zameen ki har cheez ko apni taraf kheenchne wali taqat.
- **Energy:** kaam karne ki salahiyat.
- **Conductor:** woh cheez jis mein se garmi ya bijli asani se guzre - taanba.
- **Insulator:** woh cheez jo garmi ya bijli ko roke - plastic, rabar.
- **Reflection:** roshni ya awaaz ka takra kar wapas ana.
- **Refraction:** ek cheez se doosri mein jate hue roshni ka jhukna.
- **Vibration:** tharathrahat - har awaaz isi se paida hoti hai.
- **Circuit:** bijli ke behne ka band aur mukammal rasta.
- **Eclipse:** grahan - jab koi falaki jism doosre ke saaye mein chhup jaye.
- **Satellite:** woh jism jo kisi sayyare ke gird ghumta hai - chaand zameen ka satellite hai.
- **Pollution:** mahol ka ganda hona - hawa, pani ya mitti ka.
- **Balanced Diet:** woh khana jismein tamam ajza munasib miqdar mein hon.

Revision

# Golden Facts Chart

Test se pehle ye chart ek nazar zaroor dekhein - units, ijaadat aur sab se bare records ek jagah.

- Insani jism: baligh mein 206 haddiyan; sab se lambi haddi femur, sab se chhoti stapes (kaan mein).
- Sehatmand jism ka darja hararat 37 degree C (98.6 degree F); nabz taqreeban 72 dafa per minute.
- Pani 0 degree C par jamta hai aur 100 degree C par ubalta hai.
- Hawa ki tarkeeb: Nitrogen taqreeban 78 feesad, Oxygen taqreeban 21 feesad.
- Zameen ki satah ka taqreeban 71 feesad hissa pani se dhaka hai.
- Roshni ki raftaar taqreeban 300,000 km per second - qudrat ki sab se tez raftaar.
- Awaaz ki raftaar hawa mein taqreeban 340 metre per second.
- Sab se bara sayyara Jupiter; sooraj ke sab se qareeb Mercury; sab se garam aur chamakdar Venus; laal sayyara Mars.
- Sooraj zameen ke sab se qareeb ka tara (star) hai; chaand zameen se taqreeban 384,000 km door hai.
- Duniya ka sab se bara janwar blue whale hai; sab se bara parinda shutarmurgh (ostrich) hai jo ur nahi sakta.
- Zameen apne gird 24 ghante mein aur sooraj ke gird 365¼ din mein chakkar poora karti hai.
- Ijaadat: bijli ka bulb - Thomas Edison; telephone - Alexander Graham Bell; penicillin (pehla antibiotic) - Alexander Fleming; hawai jahaz - Wright Brothers; radio - Marconi.
- Aalaat: thermometer temperature napta hai; microscope chhoti cheezein bari dikhata hai; telescope door ke sitaray dikhata hai.
- Zameen ke gird ozone ki teh sooraj ki nuqsandeh kirnon se hifazat karti hai.

## Exam Tips

- Pehle woh sawal hal karein jo pakke aate hon - mushkil sawal par waqt zaya na karein, nishan laga kar aage barhein.
- MCQ mein do ghalat options pehle kaat dein (elimination) - is se sahi jawab tak pohanchna asaan ho jata hai.
- Numbers wale sawal (bones 206, oxygen 21 feesad, pani 100 degree C) ratta lagane ki bajaye kisi cheez se jor kar yaad karein - jaise 206 haddiyan aur 32 daant.
- Science ke sawal mein roz ki zindagi ki misaal sochiye - aksar sahi jawab wohi hota hai jo aap roz dekhte hain (chamcha garam hona, bijli chamak pehle dikhna).
- Confuse karne wale jore hamesha sath yaad karein - conduction/convection, evaporation/condensation, rotation/revolution, producer/consumer.
- Diagram ya tajurbe (experiment) wale sawal mein pehle dekhein ke kya badla gaya aur kya naapa gaya - aadha jawab isi mein chhupa hota hai.
- Galat samjhute hue concepts par khaas dhyaan dein - mausam jhukao se bante hain faasle se nahi; hawa mein sab se zyada nitrogen hai oxygen nahi.

- Har chapter ke aakhir ki Answer Key se apne jawab check karein aur galat sawal copy par note karke hafte mein ek dafa dobara karein.
- Test se ek raat pehle is book ka Golden Facts wala chart ek nazar zaroor dohra lein - aksar sawal inhi lines se bante hain.
- Safai, pani aur sehat wale sawal aam tor par seedhe hote hain - in mein mushkil option ki bajaye aam feham (common sense) wala jawab chunein.

**Best of luck!** Mehnat kabhi zaya nahi jati - roz ka thora sa parhna, kal ki bari kamyabi banta hai. - MR Academy